

Exercise 10: Implementing the MVC Pattern

CODE:

// Model Class

```
class Student {  
    private String name;  
    private int id;  
    private String grade;  
    Student(String name, int id, String grade) {  
        this.name = name;  
        this.id = id;  
        this.grade = grade;  
    }  
    public String getName() {  
        return name;  
    }  
    public int getId() {  
        return id;  
    }  
    public String getGrade() {  
        return grade;  
    }  
    public void setName(String name) {  
        this.name = name;  
    }  
    public void setId(int id) {  
        this.id = id;  
    }  
    public void setGrade(String grade) {  
        this.grade = grade;  
    }  
}
```

```

    }
}

// View Class

class StudentView {

    public void displayStudentDetails(Student student) {

        System.out.println("Student Details");

        System.out.println("-----");

        System.out.println("Name : " + student.getName());

        System.out.println("ID   : " + student.getId());

        System.out.println("Grade : " + student.getGrade());

    }

}

```

```

// Controller Class

class StudentController {

    private Student model;

    private StudentView view;

    StudentController(Student model, StudentView view) {

        this.model = model;

        this.view = view;

    }

    public void setStudentName(String name) {

        model.setName(name);

    }

    public void setStudentId(int id) {

        model.setId(id);

    }

    public void setStudentGrade(String grade) {

        model.setGrade(grade);

    }

}

```

```
        public void updateView() {  
            view.displayStudentDetails(model);  
        }  
    }  
  
    // Main Class  
  
    public class Main {  
  
        public static void main(String[] args) {  
  
            Student student = new Student("Rahul", 101, "A");  
  
            StudentView view = new StudentView();  
  
            StudentController controller = new StudentController(student, view);  
  
            System.out.println("Initial Student Record");  
  
            controller.updateView();  
  
            controller.setStudentName("Sneha");  
  
            controller.setStudentId(102);  
  
            controller.setStudentGrade("A+");  
  
            System.out.println("\nUpdated Student Record");  
  
            controller.updateView();  
  
        }  
    }  
}
```

OUTPUT:

```
Initial Student Record  
Student Details  
-----  
Name   : Rahul  
ID      : 101  
Grade   : A  
  
Updated Student Record  
Student Details  
-----  
Name   : Sneha  
ID      : 102  
Grade   : A+
```